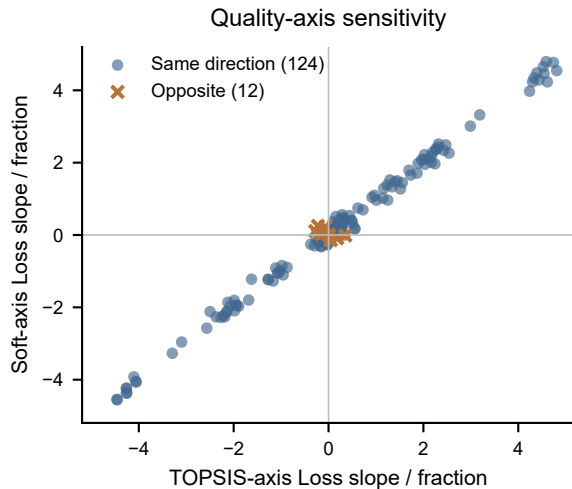
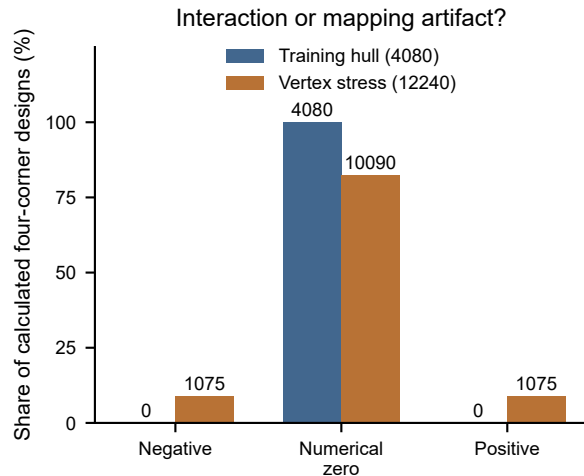


Frozen-model diagnostics, not empirical complementarity tests

a



b



Across active strength assumptions: 107/136 pairs reverse; 27/136 retain a strict direction.

Every nonzero vertex-stress interaction is explained by quality clipping; zero tolerance = $1e-10$ Loss.